1. A function f(x) is said to be periodic with period T > 0 if for all x:

(A) f(x + T) = f(x)
(B) f(x + T) = -f(x)
(C) f(x * T) = f(x)
(D) f(x + T) = T * f(x)

2. The fundamental period of sin(nx) is:

(A) 2*pi
(B) 2*pi / n
(C) pi / n
(D) n * pi

3. The fundamental period of cos(2*pi*x / L) is:

(A) L
(B) 2*L
(C) L / 2
(D) pi * L

4. Dirichlet's conditions ensure that a periodic function f(x) can be expanded in a convergent Fourier series. Which is NOT a Dirichlet condition?

(A) f(x) is single-valued and bounded
(B) f(x) has a finite number of discontinuities in any period
(C) f(x) has infinitely many local maxima and minima
(D) f(x) has a finite number of extrema in any period

5. At a point of finite discontinuity x0, the Fourier series of f(x) converges to:

(A) f(x0)
(B) (1/2) * [f(x0+) + f(x0-)]
(C) 0
(D) Infinity

6. In the Fourier series f(x) = a0/2 + sum [an cos(nx) + bn sin(nx)] on [-pi, pi], the constant term a0 is:

(A) (1/pi) * int_{-pi}^pi f(x) dx
(B) (2/pi) * int_0^pi f(x) dx
(C) (1/2pi) * int_{-pi}^pi f(x) dx
(D) int_{-pi}^pi f(x) dx

7. In Euler's formulas over [-pi, pi], the Fourier cosine coefficient an is given by:

(A) (1/pi) * int_{-pi}^pi f(x) * cos(nx) dx
(B) (1/2pi) * int_{-pi}^pi f(x) * cos(nx) dx
(C) (2/pi) * int_0^pi f(x) dx
(D) int_{-pi}^pi f(x) * sin(nx) dx

8. In Euler's formulas over [-pi, pi], the Fourier sine coefficient bn is given by:

(A) (1/pi) * int_{-pi}^pi f(x) * sin(nx) dx
(B) (1/pi) * int_{-pi}^pi f(x) * cos(nx) dx
(C) (2/pi) * int_0^pi f(x) dx
(D) 0

9. A function f(x) is called an even function if for all x:

(A) f(-x) = -f(x)
(B) f(-x) = f(x)
(C) f(x) = 0
(D) f(x + pi) = f(x)

10. A function f(x) is called an odd function if for all x:

(A) f(-x) = f(x)
(B) f(-x) = -f(x)
(C) f(x) = 1
(D) f'(x) = 0

11. The Fourier series of a real even periodic function on [-pi, pi] contains:

(A) Only sine terms
(B) Only cosine terms and constant term a0/2
(C) Both sine and cosine terms
(D) No constant term

12. For an even function f(x) on [-pi, pi], the Fourier sine coefficients bn are identically:

(A) 1
(B) 0
(C) -1
(D) pi

13. The Fourier series of a real odd periodic function on [-pi, pi] contains:

(A) Only cosine terms
(B) Only sine terms
(C) A non-zero constant term
(D) Both sine and cosine terms

14. For an odd function f(x) on [-pi, pi], the Fourier coefficients a0 and an are identically:

(A) 1
(B) 0
(C) pi
(D) n

15. The product of two odd functions is always:
- **(A)** An odd function
- **(B)** An even function
- **(C)** Neither even nor odd
- **(D)** Zero

16. The product of an even function and an odd function is always:
- **(A)** An even function
- **(B)** An odd function
- **(C)** Constant
- **(D)** Positive

17. For an arbitrary interval [-L, L], the Fourier series is expanded in terms of:
- **(A)** cos(nx) and sin(nx)
- **(B)** cos(n*pi*x / L) and sin(n*pi*x / L)
- **(C)** cos(2*n*x) and sin(2*n*x)
- **(D)** e^(nx)

18. For an arbitrary interval [-L, L], the coefficient an is defined as:
- **(A)** (1/L) * int_{-L}^L f(x) * cos(n*pi*x / L) dx
- **(B)** (1/pi) * int_{-L}^L f(x) dx
- **(C)** (2/L) * int_{-L}^L f(x) dx
- **(D)** (1/2L) * int_{-L}^L f(x) dx

19. A half-range Fourier cosine series of f(x) defined on [0, L] contains only:
- **(A)** Sine terms
- **(B)** Cosine terms including constant term a0/2
- **(C)** Odd harmonics only
- **(D)** Exponential terms

20. The coefficient an in the half-range cosine series on [0, L] is given by:
- **(A)** (1/L) * int_0^L f(x) cos(n*pi*x / L) dx
- **(B)** (2/L) * int_0^L f(x) cos(n*pi*x / L) dx
- **(C)** (1/2L) * int_0^L f(x) dx
- **(D)** (4/L) * int_0^L f(x) dx

21. A half-range Fourier sine series of f(x) on [0, L] contains:
- **(A)** Only cosine terms
- **(B)** Only sine terms (bn)
- **(C)** Both a0 and bn
- **(D)** Even powers of x

22. The coefficient bn in the half-range sine series on [0, L] is given by:
- **(A)** (1/L) * int_0^L f(x) sin(n*pi*x / L) dx
- **(B)** (2/L) * int_0^L f(x) sin(n*pi*x / L) dx
- **(C)** (2/pi) * int_0^L f(x) dx
- **(D)** 0

23. Parseval's identity for Fourier series on [-L, L] is an expression of:
- **(A)** Conservation of energy / power
- **(B)** Conservation of charge
- **(C)** Conservation of momentum
- **(D)** Newton's second law

24. Parseval's identity states that (1/L) * int_{-L}^L [f(x)]^2 dx equals:
- **(A)** a0^2 / 2 + sum (an^2 + bn^2)
- **(B)** a0^2 / 4 + sum (an^2 + bn^2)
- **(C)** sum (an + bn)
- **(D)** a0 + sum an

25. Harmonic analysis is the process of computing Fourier coefficients from:
- **(A)** Differential equations
- **(B)** Tabular / experimental discrete data values
- **(C)** Laplace transforms
- **(D)** Matrix eigenvalues

26. In practical harmonic analysis with 2m equidistant points, the constant term a0 is equal to:
- **(A)** Mean of y
- **(B)** 2 * (Mean of y)
- **(C)** (1/2) * (Mean of y)
- **(D)** Sum of y

27. In practical harmonic analysis, the first cosine harmonic coefficient a1 is given by:
- **(A)** Mean of (y * cos x)
- **(B)** 2 * Mean of (y * cos x)
- **(C)** (1/2) * Mean of (y * cos x)
- **(D)** Sum of (y * cos x)

28. In practical harmonic analysis, the first sine harmonic coefficient b1 is given by:
- **(A)** Mean of (y * sin x)
- **(B)** 2 * Mean of (y * sin x)
- **(C)** Sum of (y * sin x)
- **(D)** 0

29. The complex (exponential) form of Fourier series of f(x) on [-L, L] is:

(A) sum_{n=-infinity}^infinity cn * e^(i*n*pi*x / L)
(B) sum an * e^(nx)
(C) sum bn * e^(-nx)
(D) c0 + sum cn * cos(nx)

30. In complex Fourier series, the complex coefficient cn is given by:

(A) (1 / (2*L)) * int_{-L}^L f(x) * e^(-i*n*pi*x / L) dx
(B) (1 / L) * int_{-L}^L f(x) * e^(i*n*pi*x / L) dx
(C) (1 / pi) * int_{-L}^L f(x) dx
(D) an - i*bn

31. The relation between complex coefficients cn, c_{-n} and real coefficients an, bn is:

(A) cn = (an - i*bn) / 2 and c_{-n} = (an + i*bn) / 2
(B) cn = an + i*bn
(C) cn = an / 2
(D) cn = bn / 2

32. The value of c0 in the complex Fourier series is:

(A) a0
(B) a0 / 2
(C) 2 * a0
(D) 0

33. In biological systems, circadian rhythms oscillate with an approximate period of:

(A) 1 hour
(B) 12 hours
(C) 24 hours
(D) 28 days

34. An Electrocardiogram (ECG) waveform represents periodic electrical activity of the:

(A) Brain
(B) Heart
(C) Liver
(D) Kidney

35. The fundamental component of a periodic biomedical signal corresponds to the harmonic with:

(A) n = 0
(B) n = 1 (fundamental frequency omega0)
(C) n = 2
(D) n = infinity

36. Harmonics with n = 2, 3, 4, ... in a Fourier series are called:

(A) Subharmonics
(B) Overtone harmonics
(C) Fundamental modes
(D) DC offsets

37. Gibbs phenomenon in Fourier series describes an overshoot near:

(A) Points of continuity
(B) Points of jump discontinuity
(C) Local maxima
(D) Origin only

38. The percentage overshoot of the Gibbs phenomenon near a jump discontinuity is approximately:

(A) 4%
(B) 9% (approx 8.95%)
(C) 18%
(D) 25%

39. The integral int_{-pi}^pi cos(mx) * cos(nx) dx for positive integers m != n is:

(A) pi
(B) 2*pi
(C) 0 (orthogonality)
(D) pi / 2

40. The integral int_{-pi}^pi sin(mx) * sin(nx) dx for positive integers m = n is:

(A) 0
(B) pi
(C) 2*pi
(D) pi / 2

41. The Fourier series of f(x) = x on [-pi, pi] contains only sine terms because f(x) is:

(A) Periodic
(B) An odd function
(C) An even function
(D) Discontinuous

42. In the Fourier series of f(x) = x on [-pi, pi], the coefficient bn is given by:

(A) 2 * (-1)^(n+1) / n
(B) 2 * (-1)^n / n
(C) 1 / n
(D) 0

43. In the Fourier series of f(x) = x^2 on [-pi, pi], the coefficient a0 is:

(A) pi^2 / 3
(B) 2*pi^2 / 3
(C) pi^2
(D) 4*pi^2 / 3

44. In the Fourier series of f(x) = x^2 on [-pi, pi], the coefficient an is:

(A) 4 * (-1)^n / n^2
(B) (-1)^n / n^2
(C) 4 / n^2
(D) 0

45. Setting x = 0 in the Fourier series of f(x) = x^2 on [-pi, pi] establishes the famous sum: sum_{n=1}^infinity (-1)^(n+1) / n^2 equal to:

(A) pi^2 / 6
(B) pi^2 / 12
(C) pi^2 / 8
(D) pi^2 / 4

46. Setting x = pi in the Fourier series of f(x) = x^2 on [-pi, pi] establishes the Basel problem: sum_{n=1}^infinity 1 / n^2 equal to:

(A) pi^2 / 6
(B) pi^2 / 8
(C) pi^2 / 12
(D) pi^2 / 4

47. The Fourier series of the square wave f(x) = -k on (-pi, 0) and f(x) = +k on (0, pi) contains:

(A) Only even sine harmonics
(B) Only odd sine harmonics: (4k/pi) * sum sin((2n-1)x)/(2n-1)
(C) Only cosine harmonics
(D) Constant a0 = k

48. Evaluating the square wave Fourier series at x = pi/2 yields Leibniz's series: 1 - 1/3 + 1/5 - 1/7 + ... equal to:

(A) pi / 2
(B) pi / 4
(C) pi / 6
(D) pi / 8

49. For f(x) = |x| on [-pi, pi], f(x) is an even function. Its Fourier series has bn equal to:

(A) 0
(B) 2 / n
(C) (-1)^n / n
(D) pi / n

50. In the Fourier expansion of f(x) = |x| on [-pi, pi], the coefficient a0 is:

(A) pi
(B) pi / 2
(C) 2*pi
(D) pi^2 / 2

51. The Fourier series of f(x) = |x| on [-pi, pi] yields the sum 1/1^2 + 1/3^2 + 1/5^2 + ... equal to:

(A) pi^2 / 6
(B) pi^2 / 8
(C) pi^2 / 12
(D) pi^2 / 4

52. Find the Fourier coefficient bn for f(x) = x * cos(x) on [-pi, pi]:

(A) b1 = -1/2, bn = (-1)^n * 2n / (n^2 - 1) for n >= 2
(B) 0
(C) 1 / n
(D) (-1)^n / n^2

53. In the half-range sine series of f(x) = 1 on [0, pi], the coefficient bn is:

(A) (4 / (n*pi)) for odd n, 0 for even n
(B) 2 / (n*pi)
(C) 0
(D) (-1)^n / n

54. The half-range cosine series of f(x) = x on [0, pi] has a0 equal to:

(A) pi
(B) pi / 2
(C) 2*pi
(D) pi^2

55. In the half-range cosine series of f(x) = x on [0, pi], the coefficient an for even n (n = 2, 4, ...) is:

(A) 0
(B) -4 / (pi * n^2)
(C) 2 / n^2
(D) pi / n

56. In the half-range cosine series of f(x) = x on [0, pi], an for odd n is:

(A) -4 / (pi * n^2)
(B) 0
(C) 4 / (pi * n^2)
(D) -2 / n^2

57. Find the Fourier series of f(x) = e^x on (-pi, pi). The coefficient a0 is:

(A) (2 / pi) * sinh(pi)
(B) (1 / pi) * sinh(pi)
(C) cosh(pi)
(D) e^pi - e^(-pi)

58. In the Fourier series of f(x) = e^x on (-pi, pi), the coefficient an is:

(A) [(2 / pi) * sinh(pi)] * [(-1)^n / (1 + n^2)]
(B) [(2 / pi) * sinh(pi)] * [(-1)^n * n / (1 + n^2)]
(C) 1 / (1 + n^2)
(D) 0

59. In the Fourier series of f(x) = e^x on (-pi, pi), the coefficient bn is:

(A) [(2 / pi) * sinh(pi)] * [(-1)^(n+1) * n / (1 + n^2)]
(B) (-1)^n / (1 + n^2)
(C) 0
(D) n / (1 + n^2)

60. Apply Parseval's identity to f(x) = x^2 on [-pi, pi] to evaluate sum_{n=1}^infinity 1 / n^4:

(A) pi^4 / 90
(B) pi^4 / 45
(C) pi^4 / 96
(D) pi^4 / 120

61. Apply Parseval's identity to f(x) = x on [-pi, pi] to evaluate sum_{n=1}^infinity 1 / n^2:

(A) pi^2 / 6
(B) pi^2 / 12
(C) pi^2 / 8
(D) pi^2 / 4

62. Apply Parseval's identity to the square wave f(x) to evaluate sum_{n=1}^infinity 1 / (2n - 1)^2:

(A) pi^2 / 8
(B) pi^2 / 6
(C) pi^2 / 12
(D) pi^2 / 16

63. A half-wave rectifier outputs f(t) = I0*sin(omega*t) for 0 < omega*t < pi and 0 for pi < omega*t < 2pi. The DC average value a0/2 is:

(A) I0 / pi
(B) 2*I0 / pi
(C) I0 / 2
(D) I0 / sqrt(2)

64. A full-wave rectifier outputs f(t) = I0 * |sin(omega*t)| with period T = pi / omega. The DC average value is:

(A) 2*I0 / pi
(B) I0 / pi
(C) I0 / 2
(D) 4*I0 / pi

65. In a full-wave rectifier Fourier series, which harmonics are present?

(A) Only even cosine harmonics: - (4*I0/pi) * sum cos(2n*omega*t) / (4n^2 - 1)
(B) Only odd sine harmonics
(C) All integer harmonics
(D) No harmonics

66. The Root Mean Square (RMS) value of a periodic function f(t) with period T is defined as:

(A) sqrt((1/T) * int_0^T [f(t)]^2 dt)
(B) (1/T) * int_0^T f(t) dt
(C) (1/T) * int_0^T |f(t)| dt
(D) max |f(t)|

67. By Parseval's theorem, the RMS value of f(x) = a0/2 + sum [an cos(nx) + bn sin(nx)] is:

(A) sqrt(a0^2 / 4 + (1/2)*sum (an^2 + bn^2))
(B) a0/2 + sum (an + bn)
(C) sqrt(sum (an^2 + bn^2))
(D) a0 / sqrt(2)

68. The RMS value of a pure sinusoidal waveform x(t) = A * sin(omega*t) is:

(A) A / sqrt(2)
(B) A / 2
(C) A
(D) 2 * A / pi

69. An Electroencephalogram (EEG) signal decomposed into Fourier frequency bands includes Alpha waves in the frequency range:

(A) 8 to 13 Hz
(B) 0.5 to 4 Hz (Delta)
(C) 4 to 8 Hz (Theta)
(D) 13 to 30 Hz (Beta)

70. In discrete harmonic analysis of a biological rhythm sampled at 12 equidistant points y0, y1, ..., y11 over 24 hours, the fundamental period corresponds to N equal to:

(A) 12
(B) 6
(C) 24
(D) 2

71. The Fourier series of f(x) = cosh(x) on [-pi, pi] contains:

(A) Only cosine terms because cosh(x) is even
(B) Only sine terms
(C) Both sine and cosine terms
(D) Zero constant term

72. The Fourier series of f(x) = sinh(x) on [-pi, pi] contains:

(A) Only sine terms because sinh(x) is odd
(B) Only cosine terms
(C) A non-zero a0
(D) Zero sine terms

73. If f(x) is periodic with period 2L and f(x + L) = -f(x) (half-wave anti-symmetry), its Fourier series contains:

(A) Only odd harmonics (n = 1, 3, 5, ...)
(B) Only even harmonics
(C) Only constant term
(D) All harmonics

74. If f(x) is periodic with period 2L and f(x + L) = f(x) (half-wave symmetry), its Fourier series contains:

(A) Only even harmonics (n = 2, 4, 6, ...)
(B) Only odd harmonics
(C) No harmonics
(D) Only sine terms

75. The complex Fourier coefficient c_{-n} for a real-valued signal f(t) is related to cn by:

(A) c_{-n} = cn* (complex conjugate)
(B) c_{-n} = -cn
(C) c_{-n} = cn
(D) c_{-n} = 0

76. Find the Fourier series of f(x) = x^3 on [-pi, pi]. The coefficient bn is:

(A) 2 * (-1)^n * [(6 - n^2*pi^2) / n^3]
(B) 2 * (-1)^n / n^3
(C) (-1)^n * 6 / n^3
(D) 0

77. In the Fourier series of f(x) = x^3 on [-pi, pi], the coefficients a0 and an are:

(A) 0
(B) pi^3
(C) (-1)^n / n^2
(D) 2

78. Evaluate the Fourier series of f(x) = x*(pi - x) on [0, pi] in a half-range sine series. The coefficient bn is:

(A) 8 / (pi * n^3) for odd n, 0 for even n
(B) 4 / (pi * n^3)
(C) (-1)^n / n^3
(D) 0

79. Using the half-range sine series of f(x) = x*(pi - x), setting x = pi/2 evaluates the sum 1/1^3 - 1/3^3 + 1/5^3 - 1/7^3 + ... equal to:

(A) pi^3 / 32
(B) pi^3 / 16
(C) pi^3 / 24
(D) pi^3 / 8

80. Find the Fourier series of f(x) = |sin x| on [-pi, pi]. The coefficient a0 is:

(A) 4 / pi
(B) 2 / pi
(C) pi / 2
(D) 1

81. In the Fourier series of f(x) = |sin x| on [-pi, pi], the coefficient an is:

(A) -4 / [pi * (n^2 - 1)] for even n, 0 for odd n
(B) -2 / [pi * (n^2 - 1)]
(C) 1 / (n^2 - 1)
(D) 0

82. From the expansion of |sin x|, setting x = 0 yields the sum 1 / (1 * 3) + 1 / (3 * 5) + 1 / (5 * 7) + ... equal to:

(A) 1/2
(B) 1/4
(C) pi / 4
(D) 1

83. Find the half-range cosine series of f(x) = sin x on [0, pi]. The coefficient a0 is:

(A) 4 / pi
(B) 2 / pi
(C) 0
(D) 1

84. In the half-range cosine series of f(x) = sin x on [0, pi], the coefficient an for even n is:

(A) -4 / [pi * (n^2 - 1)]
(B) 2 / [pi * (n^2 - 1)]
(C) 0
(D) 1 / n

85. Apply Parseval's identity to the half-range sine series of f(x) = 1 on [0, pi] to evaluate sum_{n=1}^infinity 1 / (2n - 1)^2:

(A) pi^2 / 8
(B) pi^2 / 6
(C) pi^2 / 12
(D) pi^2 / 4

86. Apply Parseval's identity to the half-range sine series of f(x) = x*(pi - x) on [0, pi] to evaluate sum_{n=1}^infinity 1 / (2n - 1)^6:

(A) pi^6 / 960
(B) pi^6 / 480
(C) pi^6 / 120
(D) pi^6 / 1440

87. Find the Fourier expansion of f(x) = x^2 on [-L, L]. The coefficient an is:

(A) 4*L^2 * (-1)^n / (n^2 * pi^2)
(B) 2*L^2 / (n^2 * pi^2)
(C) (-1)^n / n^2
(D) 0

88. In the Fourier expansion of f(x) = x^2 on [-L, L], the constant term a0/2 is:

(A) L^2 / 3
(B) 2*L^2 / 3
(C) L^2 / 6
(D) L^2

89. Find the complex Fourier coefficient cn for f(x) = e^(a*x) on [-pi, pi]:

(A) [sinh(a*pi) / pi] * [(-1)^n / (a - i*n)]
(B) 1 / (a - i*n)
(C) [(-1)^n / pi] * sinh(a*pi)
(D) 0

90. In practical harmonic analysis, given 6 values of y: y0=9, y1=18, y2=24, y3=28, y4=26, y5=20 at x = 0, 60, 120, 180, 240, 300 degrees. The constant term a0/2 (mean of y) is:

(A) 20.83
(B) 25.0
(C) 18.5
(D) 22.0

91. For the data in Q90, the first harmonic cosine coefficient a1 = (2/6)*sum(y*cos x) evaluates approximately to:

(A) -5.0
(B) -2.5
(C) 3.0
(D) 0

92. For the data in Q90, the first harmonic sine coefficient b1 = (2/6)*sum(y*sin x) evaluates approximately to:

(A) -3.46
(B) 4.5
(C) 1.2
(D) -1.5

93. An ECG signal is approximated by f(t) = a0/2 + sum [an cos(n*omega0*t) + bn sin(n*omega0*t)]. If heart rate is 72 beats/minute, the fundamental frequency f0 is:

(A) 1.2 Hz
(B) 72 Hz
(C) 0.83 Hz
(D) 2.4 Hz

94. In Q93, the fundamental angular frequency omega0 is:

(A) 2.4 * pi rad/s = 7.54 rad/s
(B) 1.2 rad/s
(C) 72 rad/s
(D) 144*pi rad/s

95. The 5th harmonic frequency of the ECG signal with heart rate 72 bpm is:

(A) 6.0 Hz
(B) 1.2 Hz
(C) 5.0 Hz
(D) 10.0 Hz

96. A high-pass filter is designed to remove baseline wander (< 0.5 Hz) from an ECG signal while preserving the QRS complex. The QRS complex predominantly contains Fourier frequencies in the range:

(A) 10 to 40 Hz
(B) 0.01 to 0.1 Hz
(C) 100 to 500 Hz
(D) 1 to 2 kHz

97. Total Harmonic Distortion (THD) of a biological signal with fundamental amplitude V1 and harmonic amplitudes V2, V3, ... is defined as:

(A) sqrt(sum_{n=2}^infinity Vn^2) / V1
(B) sum Vn / V1
(C) V1 / V2
(D) V2^2 / V1^2

98. Evaluate the Fourier series of f(x) = x^4 on [-pi, pi]. The sum sum_{n=1}^infinity 1 / n^4 deduced from it without Parseval's identity is:

(A) pi^4 / 90
(B) pi^4 / 45
(C) pi^4 / 96
(D) pi^4 / 120

99. Evaluate the Fourier series of f(x) = x^4 on [-pi, pi]. The alternating sum sum_{n=1}^infinity (-1)^(n+1) / n^4 is:

(A) 7 * pi^4 / 720
(B) pi^4 / 90
(C) pi^4 / 180
(D) 3 * pi^4 / 720

100. If a biological enzyme secretion rate oscillates as f(t) = 5 + 3*cos(2*pi*t/24) + 1.5*sin(2*pi*t/24) (t in hours), the amplitude of the circadian oscillation is:

(A) sqrt(3^2 + 1.5^2) = sqrt(11.25) ~ 3.35
(B) 4.5
(C) 5.0
(D) 3.0

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State Dirichlet's Conditions for the expansion of a periodic function f(x) in a Fourier series. State the convergence theorem for continuous points and jump discontinuities.

Q2. Prove the Orthogonality Relations of the trigonometric system {1, cos(nx), sin(nx)} on the interval [-pi, pi] for all positive integers m and n.

Q3. Derive Euler's Formulas for the Fourier coefficients a0, an, and bn of a periodic function f(x) on the interval [-pi, pi] using orthogonality relations.

Q4. Find the Fourier series of the square wave: f(x) = -1 for -pi < x < 0, and f(x) = +1 for 0 < x < pi. Hence, deduce Leibniz's Gregory series for pi/4.

Q5. Prove that if f(x) is an even function on [-pi, pi], its Fourier series reduces to a Fourier Cosine Series with bn = 0, and if f(x) is an odd function, it reduces to a Fourier Sine Series with a0 = an = 0.

Q6. Find the Fourier series of f(x) = x on the interval [-pi, pi]. State its value at points of discontinuity and deduce that 1 - 1/3 + 1/5 - 1/7 + ... = pi/4.

Q7. Find the Fourier series of f(x) = x^2 on [-pi, pi]. Hence, deduce: (i) sum_{n=1}^infinity 1/n^2 = pi^2/6 (Euler's Basel Problem), (ii) sum_{n=1}^infinity (-1)^(n+1)/n^2 = pi^2/12, and (iii) sum_{n=1}^infinity 1/(2n-1)^2 = pi^2/8.

Q8. Find the Fourier series of f(x) = |x| on [-pi, pi]. Hence, deduce that: 1/1^2 + 1/3^2 + 1/5^2 + ... = pi^2 / 8.

Q9. Find the Fourier series of f(x) = x + x^2 on the interval [-pi, pi]. Hence evaluate the sum: sum_{n=1}^infinity 1/n^2.

Q10. Derive the Fourier series expansion formulas for a function f(x) defined on an arbitrary symmetric interval [-L, L] by change of scale of variable.

Q11. Find the Fourier series of f(x) = x on the interval [-2, 2] (period 2L = 4, so L = 2).

Q12. Find the Fourier series of f(x) = 2x - x^2 on the interval [0, 3] with period T = 3.

Q13. Explain the concept of Half-Range Fourier Expansions. Detail how an even extension yields a Half-Range Cosine Series and an odd extension yields a Half-Range Sine Series on [0, L].

Q14. Obtain the Half-Range Sine Series of f(x) = x*(pi - x) on the interval [0, pi]. Hence, deduce the sum: 1/1^3 - 1/3^3 + 1/5^3 - 1/7^3 + ... = pi^3 / 32.

Q15. Obtain the Half-Range Cosine Series of f(x) = x*(pi - x) on the interval [0, pi]. Hence, deduce the sum: sum_{n=1}^infinity 1/n^4 = pi^4 / 90.

Q16. Obtain the Half-Range Cosine Series for f(x) = sin(x) on the interval [0, pi]. (This represents full-wave rectification in electrical/biomedical engineering).

Q17. State Parseval's Identity for a Fourier series on [-L, L]. Prove it from Euler's formulas and provide its physical interpretation in signal power and energy conservation.

Q18. Apply Parseval's Identity to the Fourier series of f(x) = x on [-pi, pi] to evaluate the sum: sum_{n=1}^infinity 1 / n^2.

Q19. Apply Parseval's Identity to the Fourier series of f(x) = x^2 on [-pi, pi] to evaluate the sum: sum_{n=1}^infinity 1 / n^4.

Q20. Apply Parseval's Identity to the half-range sine series of f(x) = 1 on [0, pi] to evaluate: sum_{n=1}^infinity 1 / (2n - 1)^2.

Q21. Derive the Complex (Exponential) Form of Fourier Series from the real trigonometric Fourier series using Euler's formula e^(i*theta) = cos(theta) + i*sin(theta).

Q22. Find the Complex Fourier Series of f(x) = e^(a*x) on [-pi, pi] (where a is a non-zero real constant).

Q23. Explain the Gibbs Phenomenon in Fourier series. Calculate the approximate overshoot near a jump discontinuity of height 2 for a square wave.

Q24. Explain Practical Harmonic Analysis. Derive formulas for computing the Fourier coefficients a0, an, and bn from a set of 2m discrete equidistant experimental data points.

Q25. Perform practical harmonic analysis on the following 6-point periodic dataset to find the constant term and the first harmonic of the Fourier series: x = [0, 60, 120, 180, 240, 300] deg, y = [0.8, 0.6, 0.4, 0.7, 0.9, 1.1].

Q26. In biotechnology and endocrinology, circadian hormone secretion (such as serum cortisol) follows a 24-hour diurnal cycle. Model cortisol concentration C(t) = a0/2 + a1*cos(pi*t/12) + b1*sin(pi*t/12). Given measurements C(0)=15 ug/dL (midnight), C(6)=25 ug/dL (6 AM), C(12)=10 ug/dL

Q27. Explain how Fourier Series decomposition is used in biomedical engineering to analyze Electrocardiogram (ECG) signals and filter out 50 Hz powerline interference.

Q28. In respiratory biotechnology, lung airflow during breathing is modeled as a periodic square/trapezoidal flow rate. Find the Fourier series for airflow V'(t) = V0 on [0, T/2] (inhalation) and V'(t) = -V0 on [T/2, T] (exhalation).

Q29. In Heart Rate Variability (HRV) analysis, explain how the Fourier power spectral density of the RR-interval tachogram is used to assess autonomic nervous system balance (LF/HF ratio).

Q30. A periodic biological bioreactor stirrer oscillation velocity is given by v(t) = t^2 on (-1, 1) with period 2 (L = 1). Find its Fourier series, and use it to determine the fraction of total kinetic energy contained in the fundamental frequency harmonic.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[B]	**Q41:**	[B]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[B]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[C]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[B]	**Q25:**	[B]	**Q45:**	[B]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[B]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[B]	**Q48:**	[B]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[B]	**Q31:**	[A]	**Q51:**	[B]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[B]	**Q33:**	[C]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[B]	**Q37:**	[B]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[B]	**Q39:**	[C]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

zero:
int_{-pi}^pi sin(mx) * cos(nx) dx = 0 for all integers m, n.
Conclusion:
The trigonometric set is mutually orthogonal on [-pi, pi].

= (2 / (n * pi)) * [-cos(n*pi) - (-cos 0)]
= (2 / (n * pi)) * [1 - (-1)^n].
Step 3: Evaluate bn for Even and Odd n:
- If n is even (n = 2, 4, 6, ...):
(-1)^n = 1 => 1 - 1 = 0 => bn = 0.
- If n is odd (n = 1, 3, 5, ..., let n = 2k - 1):
(-1)^n = -1 => 1 - (-1) = 2 => bn = (2 / (n*pi)) * 2 = 4 / (n * pi).
Step 4: Assemble the Fourier Series:
f(x) = sum_{k=1}^infinity [4 / ((2k - 1)*pi)] * sin((2k - 1)x)
= (4 / pi) * [sin x + (sin 3x) / 3 + (sin 5x) / 5 + (sin 7x) / 7 + ...].
Step 5: Deduction of Leibniz's Series:
- Substitute x = pi / 2 (a point of continuity where f(pi/2) = 1):
sin(pi / 2) = 1
sin(3*pi / 2) = -1
sin(5*pi / 2) = 1
sin(7*pi / 2) = -1.
- Substitute into the Fourier series:
1 = (4 / pi) * [1 - 1/3 + 1/5 - 1/7 + ...].
- Multiply both sides by pi / 4:
1 - 1/3 + 1/5 - 1/7 + ... = pi / 4.
Conclusion:
The Fourier series is (4/pi)*sum sin((2n-1)x)/(2n-1), and Leibniz's series pi/4 is verified.

>>Solution to Question 5: Prove that if f(x) is an even function on [-pi, pi], its Fourier series reduces to a Fourier Cosine Series with bn = 0, and if f(x) is an odd fu

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>>Solution Statement Question 8 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 9 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 10 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 11 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 12 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 13 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 14 Find the Fourier series of f(x) = x^2 in the interval Hence deduce its value at the points of discontinuity Solution Statement Question 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= pi^2 / 3 - 4 * [(cos x)/1^2 - (cos 2x)/2^2 + (cos 3x)/3^2 - (cos 4x)/4^2 + ...].
Step 5: Deduce Series (i) - Euler's Basel Problem:
Set x = pi (point of continuity where f(pi) = pi^2):
cos(n*pi) = (-1)^n.
pi^2 = pi^2 / 3 + 4 * sum_{n=1}^infinity [(-1)^n * (-1)^n / n^2] = pi^2 / 3 + 4 *
sum_{n=1}^infinity 1 / n^2
pi^2 - pi^2 / 3 = 4 * sum_{n=1}^infinity 1 / n^2
(2/3)*pi^2 = 4 * sum_{n=1}^infinity 1 / n^2
=> sum_{n=1}^infinity 1 / n^2 = (2/3)*pi^2 / 4 = pi^2 / 6. (Part i proved).
Step 6: Deduce Series (ii):
Set x = 0 (point of continuity where f(0) = 0):
0 = pi^2 / 3 + 4 * sum_{n=1}^infinity [(-1)^n / n^2]
- pi^2 / 3 = - 4 * sum_{n=1}^infinity [(-1)^(n+1) / n^2]
=> sum_{n=1}^infinity (-1)^(n+1) / n^2 = pi^2 / 12. (Part ii proved).
Step 7: Deduce Series (iii) (Sum of Odd Reciprocals):
Add series (i) and (ii):
2 * sum_{k=1}^infinity 1 / (2k - 1)^2 = pi^2 / 6 + pi^2 / 12 = 3*pi^2 / 12 = pi^2 / 4
=> sum_{k=1}^infinity 1 / (2k - 1)^2 = pi^2 / 8. (Part iii proved).

>> **Solution to Question 8:** Find the Fourier series of f(x) = |x| on [-pi, pi]. Hence, deduce that: 1/1^2 + 1/3^2 + 1/5^2 + ... = pi^2 / 8.

- Substitute t = pi*x / L and dt = (pi / L) dx:
a0 = (1 / pi) * int_{-L}^L f(x) * (pi / L) dx = (1 / L) * int_{-L}^L f(x) dx.
an = (1 / pi) * int_{-L}^L f(x) * cos(n*pi*x / L) * (pi / L) dx = (1 / L) * int_{-L}^L f(x) * cos(n*pi*x / L) dx.
bn = (1 / pi) * int_{-L}^L f(x) * sin(n*pi*x / L) * (pi / L) dx = (1 / L) * int_{-L}^L f(x) * sin(n*pi*x / L) dx.
4. Summary of Formulas for [-L, L]:
f(x) = a0 / 2 + sum_{n=1}^infinity [an * cos(n*pi*x / L) + bn * sin(n*pi*x / L)],
with:
a0 = (1 / L) * int_{-L}^L f(x) dx
an = (1 / L) * int_{-L}^L f(x) * cos(n*pi*x / L) dx
bn = (1 / L) * int_{-L}^L f(x) * sin(n*pi*x / L) dx. (Proved).

>>Solution Step Question 12: Find the Fourier series of f(x) = 2x - x^2 in the interval [0, 2] (period 2L = 4, so L = 2).

Integrate by parts:
u = 2x - x^2, u' = 2 - 2x, u'' = -2, u''' = 0.
v = cos(kx), v1 = sin(kx)/k, v2 = -cos(kx)/k^2, v3 = -sin(kx)/k^3, where k = 2*n*pi / 3.
an = (2 / 3) * [(2x - x^2)*(sin kx / k) - (2 - 2x)*(-cos kx / k^2) + (-2)*(-sin kx / k^3)
]_0^3.
Since sin(k*3) = sin(2*n*pi) = 0 and sin(0) = 0, the sine terms vanish at both limits:
an = (2 / 3) * [(2 - 2x)*cos(kx) / k^2]_0^3
= (2 / 3) * [(2 - 6)*cos(2n*pi) / k^2 - (2 - 0)*cos(0) / k^2]
= (2 / 3) * [-4(1) / k^2 - 2(1) / k^2] = (2 / 3) * [-6 / k^2] = -4 / k^2.
Substitute k = 2*n*pi / 3 => k^2 = 4*n^2*pi^2 / 9:
an = -4 / [4*n^2*pi^2 / 9] = - 9 / (n^2 * pi^2).
Step 4: Compute bn:
bn = (2 / 3) * int_0^3 (2x - x^2) * sin(kx) dx
= (2 / 3) * [(2x - x^2)*(-cos kx / k) - (2 - 2x)*(-sin kx / k^2) + (-2)*(cos kx / k^3)]_0^3
= (2 / 3) * [- (2x - x^2)*cos(kx) / k - 2*cos(kx) / k^3]_0^3.
At x = 3: -(6 - 9)(1)/k - 2(1)/k^3 = 3/k - 2/k^3.
At x = 0: -(0)(1)/k - 2(1)/k^3 = -2/k^3.
Difference = (3/k - 2/k^3) - (-2/k^3) = 3 / k.
bn = (2 / 3) * [3 / k] = 2 / k = 2 / (2*n*pi / 3) = 3 / (n * pi).
Step 5: Assemble the Fourier Series:
f(x) = - (9 / pi^2) * sum_{n=1}^infinity [cos(2*n*pi*x / 3) / n^2] + (3 / pi) *
sum_{n=1}^infinity [sin(2*n*pi*x / 3) / n].

>>Solution to Question 18 Explain the concept of Half-Range Fourier Expansions. Detail how an even extension yields a Half-Range Cosine Series and an odd ex

[Problem Statement]: Obtain the Half-Range Sine Series of f(x) = x*(pi - x) on the interval [0, pi]. Hence, deduce the sum: 1/1^3 - 1/3^3 + 1/5^3 -
[Detailed Step-by-Step Answer]:
Step 1: Set up the Half-Range Sine Coefficient bn (L = pi):
bn = (2 / pi) * int_0^pi (pi*x - x^2) * sin(nx) dx.
Step 2: Integrate by Parts:
u = pi*x - x^2, u' = pi - 2x, u'' = -2, u''' = 0.
v = sin(nx), v1 = -cos(nx)/n, v2 = -sin(nx)/n^2, v3 = cos(nx)/n^3.
bn = (2 / pi) * [(pi*x - x^2)*(-cos nx / n) - (pi - 2x)*(-sin nx / n^2) + (-2)*(cos nx / n^3)
]_0^pi.
Step 3: Evaluate at Limits:
- First term vanishes at both limits since pi*x - x^2 = 0 at x = 0 and x = pi.
- Second term vanishes since sin(n*pi) = sin(0) = 0.
- Third term:
At x = pi: -2 * cos(n*pi) / n^3 = -2 * (-1)^n / n^3.
At x = 0: -2 * cos(0) / n^3 = -2 / n^3.
Difference = [-2*(-1)^n / n^3] - [-2 / n^3] = (2 / n^3) * [1 - (-1)^n].
bn = (2 / pi) * (2 / n^3) * [1 - (-1)^n] = (4 / (pi * n^3)) * [1 - (-1)^n].
Step 4: Values for Even and Odd n:
- If n is even: 1 - (-1)^n = 0 => bn = 0.
- If n is odd (n = 2k - 1): 1 - (-1)^n = 2 => bn = 8 / (pi * n^3).
Step 5: Assemble the Half-Range Sine Series:
x*(pi - x) = (8 / pi) * sum_{k=1}^infinity [sin((2k - 1)x) / (2k - 1)^3]
= (8 / pi) * [(sin x)/1^3 + (sin 3x)/3^3 + (sin 5x)/5^3 + ...].
Step 6: Deduction at x = pi / 2:
- At x = pi / 2:
f(pi/2) = (pi/2) * (pi - pi/2) = (pi/2) * (pi/2) = pi^2 / 4.
sin(pi/2) = 1, sin(3pi/2) = -1, sin(5pi/2) = 1, sin(7pi/2) = -1.
- Substitute into series:
pi^2 / 4 = (8 / pi) * [1/1^3 - 1/3^3 + 1/5^3 - 1/7^3 + ...].
- Multiply both sides by pi / 8:
1/1^3 - 1/3^3 + 1/5^3 - 1/7^3 + ... = (pi^2 / 4) * (pi / 8) = pi^3 / 32.
Conclusion:
The half-range sine series is established and the alternating cubic reciprocal sum evaluates to
pi^3 / 32.

>> Solution Step Question 15: Obtain the Half-Range Cosine Series of f(x) = x*(pi - x) on the interval [0, pi]. Hence, deduce the sum: sum_{n=1}^infinity 1/n^4 = pi^4 /

= (2 / pi) * [-pi * (-1)^n / n^2 - pi / n^2]
= - (2 / n^2) * [1 + (-1)^n].
Step 3: Values for Even and Odd n:
- If n is odd: 1 + (-1)^n = 0 => an = 0.
- If n is even (n = 2m): 1 + (-1)^n = 2 => an = -4 / n^2 = -4 / (2m)^2 = -1 / m^2.
Step 4: Assemble the Half-Range Cosine Series:
x*(pi - x) = pi^2 / 6 - sum_{m=1}^infinity [cos(2mx) / m^2]
= pi^2 / 6 - [(cos 2x)/1^2 + (cos 4x)/2^2 + (cos 6x)/3^2 + ...].
Step 5: Deduction of Basel Problem (sum 1/m^2):
At x = 0: f(0) = 0 => 0 = pi^2 / 6 - sum_{m=1}^infinity 1/m^2 => sum 1/m^2 = pi^2 / 6.
Step 6: Deduction of sum 1/m^4 using Parseval's Identity:
- For half-range cosine series on [0, pi]:
(2 / pi) * int_0^pi [f(x)]^2 dx = a0^2 / 2 + sum_{m=1}^infinity am^2.
- Left-hand side:
int_0^pi [x*(pi - x)]^2 dx = int_0^pi (pi^2*x^2 - 2pi*x^3 + x^4) dx
= [pi^2*x^3/3 - 2pi*x^4/4 + x^5/5]_0^pi = pi^5 * [1/3 - 1/2 + 1/5] = pi^5 / 30.
(2 / pi) * (pi^5 / 30) = pi^4 / 15.
- a0^2 / 2 = (pi^2 / 3)^2 / 2 = pi^4 / 18.
- Substitute:
pi^4 / 15 = pi^4 / 18 + sum_{m=1}^infinity (1 / m^2)^2 = pi^4 / 18 + sum 1/m^4
sum_{m=1}^infinity 1 / m^4 = pi^4 / 15 - pi^4 / 18 = (6 - 5)*pi^4 / 90 = pi^4 / 90.
Conclusion:
The half-range cosine series is derived and sum 1/m^4 = pi^4 / 90 is proven.

>>**Solution to Question 16:** Obtain the Half-Range Cosine Series for f(x) = sin(x) on the interval [0, pi]. (This represents full-wave rectification in electrical/biomedica

Step 5: Deduction at x = 0:
0 = 2 / pi - (4 / pi) * [1/(1*3) + 1/(3*5) + 1/(5*7) + ...]
=> 1/(1*3) + 1/(3*5) + 1/(5*7) + ... = (2 / pi) * (pi / 4) = 1/2.
Conclusion:
The full-wave rectification Fourier series is established, and the sum of telescopic fractions evaluates to 1/2.

>>Solution Question 18: Parseval's identity for the Fourier series of f(x) = x. Prove it and evaluate the sum. Provide its physical interpretation in signal pow

[f(x)]^2 = x^2.
int_{-pi}^pi x^2 dx = 2 * int_0^pi x^2 dx = 2 * [x^3 / 3]_0^pi = 2 * pi^3 / 3.
LHS = (1 / pi) * (2*pi^3 / 3) = 2*pi^2 / 3.
Step 4: Evaluate Right-Hand Side:
a0 = 0, an = 0.
bn^2 = [2 * (-1)^(n+1) / n]^2 = 4 / n^2.
RHS = sum_{n=1}^infinity bn^2 = sum_{n=1}^infinity 4 / n^2 = 4 * sum_{n=1}^infinity 1 / n^2.
Step 5: Equate LHS and RHS:
2*pi^2 / 3 = 4 * sum_{n=1}^infinity 1 / n^2.
Divide both sides by 4:
sum_{n=1}^infinity 1 / n^2 = (2*pi^2 / 3) / 4 = pi^2 / 6.
Conclusion:
Parseval's identity confirms Euler's Basel sum: sum_{n=1}^infinity 1/n^2 = pi^2 / 6.

>>**Solution to Question 20** Parseval's Identity to the Fourier series of f(x) = x^2 on [-pi, pi] to evaluate the sum sum_{n=1}^infinity 1/n^2.

Step 2: Parseval's Identity for Half-Range Sine Series on [0, pi]:
(2 / pi) * int_0^pi [f(x)]^2 dx = sum_{k=1}^infinity bn^2.
Step 3: Evaluate Left-Hand Side:
[f(x)]^2 = 1^2 = 1.
int_0^pi 1 dx = pi.
LHS = (2 / pi) * pi = 2.
Step 4: Evaluate Right-Hand Side:
bn^2 = [4 / ((2k - 1)*pi)]^2 = 16 / [pi^2 * (2k - 1)^2].
RHS = sum_{k=1}^infinity [16 / (pi^2 * (2k - 1)^2)] = (16 / pi^2) * sum_{k=1}^infinity 1 / (2k - 1)^2.
Step 5: Equate and Solve:
2 = (16 / pi^2) * sum_{k=1}^infinity 1 / (2k - 1)^2.
Multiply by pi^2 / 16:
sum_{k=1}^infinity 1 / (2k - 1)^2 = 2 * (pi^2 / 16) = pi^2 / 8.
Conclusion:
1/1^2 + 1/3^2 + 1/5^2 + ... = pi^2 / 8.

>>**Solution to Question 22:** The Complex Fourier Series of f(x) = e^(ax) is ... (where a is a non-zero real constant). Express e^(i*theta) using Euler's formula e^(i*theta) = c

Step 2: Evaluate the Integral:
cn = (1 / (2*pi)) * [e^((a - i*n)*x) / (a - i*n)]_{-pi}^pi
= [1 / (2*pi * (a - i*n))] * [e^((a - i*n)*pi) - e^(-(a - i*n)*pi)].
Step 3: Simplify Exponentials:
e^((a - i*n)*pi) = e^(a*pi) * e^(-i*n*pi) = e^(a*pi) * (-1)^n.
e^(-(a - i*n)*pi) = e^(-a*pi) * e^(i*n*pi) = e^(-a*pi) * (-1)^n.
Difference = (-1)^n * [e^(a*pi) - e^(-a*pi)] = (-1)^n * 2 * sinh(a*pi).
Step 4: Substitute back into cn:
cn = [1 / (2*pi * (a - i*n))] * [(-1)^n * 2 * sinh(a*pi)]
= [sinh(a*pi) / pi] * [(-1)^n / (a - i*n)].
Step 5: Rationalize the Denominator:
1 / (a - i*n) = (a + i*n) / (a^2 + n^2).
cn = [sinh(a*pi) / pi] * (-1)^n * [(a + i*n) / (a^2 + n^2)].
Step 6: Assemble the Complex Fourier Series:
e^(ax) = [sinh(a*pi) / pi] * sum_{n=-infinity}^infinity [(-1)^n * (a + i*n) / (a^2 + n^2)]
* e^(i*nx).
Conclusion:
The complex Fourier series of e^(ax) is established.

>> Solution to Question 23

Explain the Gibbs Phenomenon in Fourier series. Calculate the approximate overshoot near a jump discontinuity of height 2 for a squar

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>>Solution Question 28 Practical Harmonic Analysis. Derive formulas for complex Fourier coefficients and the first set of Fourier

Sum of (y * cos x) = 0.8000 + 0.3000 - 0.2000 - 0.7000 - 0.4500 + 0.5500 = 0.3000.
a1 = (2 / N) * sum(y * cos x) = (2 / 6) * 0.3000 = (1 / 3) * 0.3000 = 0.1000.
Step 4: Calculate First Harmonic Sine Coefficient b1:
Sum of (y * sin x) = 0.0000 + 0.5196 + 0.3464 + 0.0000 - 0.7794 - 0.9526 = -0.8660.
b1 = (2 / N) * sum(y * sin x) = (2 / 6) * (-0.8660) = -0.2887.
Step 5: Amplitude and Phase of First Harmonic:
Amplitude C1 = sqrt(a1^2 + b1^2) = sqrt((0.1)^2 + (-0.2887)^2) = sqrt(0.01 + 0.0833) =
sqrt(0.0933) = 0.3055.
Phase angle phi1 = arctan(a1 / b1) = arctan(0.1 / -0.2887) ~ -19.1 degrees.
Step 6: First Harmonic Fourier Approximation:
y(x) ~ 0.75 + 0.100 * cos(x) - 0.289 * sin(x)
= 0.75 + 0.306 * sin(x - 19.1 deg).
Conclusion:
The Fourier expansion up to first harmonic is y ~ 0.75 + 0.10 cos(x) - 0.29 sin(x).

>>Solution to Question 26 In biotechnology and endocrinology, circadian hormone secretion (such as serum cortisol) follows a 24-hour diurnal cycle. Model cortis

The circadian cortisol model is C(t) = 13.75 + 2.5*cos(pi*t/12) + 10*sin(pi*t/12), with peak secretion occurring at approximately 5:04 AM.

= (2 / T) * [int_0^(T/2) V0 * cos(n*omega0*t) dt - int_{T/2}^T V0 * cos(n*omega0*t) dt]
= (2*V0 / (n*omega0*T)) * [sin(n*omega0*t)]_0^(T/2) - (2*V0 / (n*omega0*T)) * [sin(n*omega0*t)]_{T/2}^T.
Since omega0*T = 2*pi, n*omega0*T/2 = n*pi, and sin(n*pi) = sin(0) = sin(2n*pi) = 0:
an = 0 for all n >= 1.
Step 5: Compute bn:
bn = (2 / T) * int_0^T V'(t) * sin(n*omega0*t) dt
= (2*V0 / T) * [int_0^(T/2) sin(n*omega0*t) dt - int_{T/2}^T sin(n*omega0*t) dt]
= [2*V0 / (n*pi)] * [-cos(n*pi) - (-cos 0) - (-cos 2n*pi - (-cos n*pi))] / 2
= [2*V0 / (n*pi)] * [1 - (-1)^n].
Step 6: Values of bn:
- For even n: bn = 0.
- For odd n (n = 2k - 1): bn = 4*V0 / ((2k - 1)*pi).
Step 7: Fourier Expansion:
V'(t) = (4*V0 / pi) * sum_{k=1}^infinity [sin((2k - 1)*2*pi*t / T) / (2k - 1)].
Conclusion:
The respiratory airflow contains only odd sinusoidal harmonics, matching the square wave Fourier series.

>> Solution to Question 29: In Heart Rate Variability (HRV) analysis, explain how the Fourier power spectral density of the RR-interval tachogram is used to assess

>> **Solution Statement Question 30:** A periodic biological bioreactor stirrer oscillation velocity is given by v(t) = t^2 on (-1, 1) with period 2 (L = 1). Find its Fourier series, and